

Generative Retrieval for Cover Song Identification via Audio-Derived Semantic IDs

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Abstract—Generative retrieval has replaced embedding search in industrial recommender systems, with TIGER and PLUM demonstrating that an autoregressive model generating discrete Semantic IDs can outperform a learned bi-encoder. We test whether this paradigm transfers to audio retrieval, using cover song identification (CSI) as a clean testbed with unambiguous ground truth. A T5 model autoregressively generates the Semantic ID of a cover given the query track’s Semantic ID. We compare four constructions of audio-derived Semantic IDs: random tuples, EnCodec RVQ codes used directly, RQ-VAE quantization of MERT-v1-95M embeddings, and RQ-VAE quantization of CLEWS (Serra et al. 2025) embeddings. Prefix-overlap analysis before any T5 training already establishes a clean ordering: random (no structure) $\ll$ EnCodec ($1.8\times$ within/cross) $\ll$ MERT ($4.3\times$) $\ll$ CLEWS ($91\times$). On Discogs-VI transductive retrieval, the same ordering dominates: MRR climbs from 0.11 (random) and 0.11 (EnCodec) through 0.27 (MERT) to 0.71 (CLEWS), with CLEWS reaching NAR 0.44, in the range of the CLEWS bi-encoder SOTA. The result confirms that generative retrieval transfers to audio retrieval, and that the quality of the source embedder, not the T5 model, is the binding constraint, exactly as TIGER and PLUM established for recommendation.

Keywords—cover song identification, generative retrieval, semantic IDs, music information retrieval, MERT, EnCodec

I. INTRODUCTION

Generative retrieval has replaced the traditional encode-and-search pipeline in industrial recommender systems: TIGER [2] showed that an autoregressive transformer generating discrete Semantic IDs outperforms a learned bi-encoder, and PLUM [18] validated the approach at industrial scale. The model’s parameters become the index. We ask: does this paradigm transfer to audio retrieval?

We use cover song identification (CSI) as a clean testbed. CSI is the task of retrieving recordings of the same underlying composition under arbitrary changes of key, tempo, instrumentation, and production [8]–[10]. Ground truth is unambiguous (two versions of “Yesterday” are positives; the rest of the corpus is not), open datasets exist at scale, and current systems are well-characterized bi-encoder retrievers [11], [12], [17]. If audio-derived Semantic IDs encode compositional structure, covers of the same composition should land in nearby regions of the SemID space and share token prefixes.

The architectural observation motivating this work is that the RQ-VAE used to construct Semantic IDs [2] and the residual vector quantizer in neural audio codecs such as EnCodec [7] are the same mechanism applied at different stages. This raises a natural question: how much does the choice of audio encoder feeding the quantizer matter? We compare four conditions spanning the spectrum of audio representation quality: random tuples (no structure), EnCodec direct codes (acoustic reconstruction features), RQ-VAE-over-MERT (general-purpose music representation), and RQ-VAE-over-CLEWS (a contrastively-trained version-invariant CSI encoder).

Our contributions are:

- The first formulation of CSI as generative retrieval over a discrete Semantic ID vocabulary, and the first demonstration that the paradigm transfers from recommendation to audio retrieval.
- A systematic comparison of four SemID constructions ranging from random tuples to RQ-VAE quantization of the current CSI SOTA encoder (CLEWS [17]), evaluated on a Discogs-VI subset of 3,413 tracks.
- A T5-CLEWS-RQ-VAE generative retriever that matches the CLEWS bi-encoder SOTA (NAR 0.44) while replacing the search index with model parameters.
- Direct evidence, via pre-training prefix-overlap analysis and post-training error analysis, that the quality of the source audio embedder is the binding constraint on generative retrieval performance, exactly as TIGER and PLUM established for recommendation.

II. RELATED WORK

A. Generative Retrieval for Recommendation

Generative retrieval replaces the encode-and-search pipeline with a single sequence model that emits a target’s identifier directly. DSI [1] cast document retrieval as text-to-identifier generation and trained with a joint indexing-and-retrieval objective. TIGER [2] introduced Semantic IDs constructed via RQ-VAE over item embeddings, with the headline finding that

structured codes substantially outperform arbitrary identifiers. PLUM [18] extended this to industrial-scale recommendation and demonstrated that the paradigm holds at catalog sizes of tens of millions. Music-domain extensions include Text2Tracks [3] (text prompt to track), FusID [4] (multimodal Semantic IDs for playlist continuation), and large-scale catalog work at industry labs [5]. All target recommendation or text-query settings.

B. Cover Song Identification

Standard benchmarks are Covers80 [10] (80 cliques, 160 tracks), Da-TACOS [9] (25k tracks), and the recent Discogs-VI [8] (493k versions across 98k cliques, with paired YouTube IDs for audio crawl). System families include similarity-matrix alignment over CQT/HPCP features and learned bi-encoders, of which MOVE [11] and CoverHunter [12] were strong recent baselines. The current SOTA is CLEWS [17], a 1024-d contrastively-trained encoder that achieves NAR around 0.43 on Discogs-VI test sets. Recent work also explores music language models (e.g. MERT [6]) as universal feature extractors.

C. The Bridge We Make

No published work has applied the generative retrieval paradigm to cover song identification, where the query and target are both audio and ground truth is unambiguous. Our work bridges the two literatures: we treat the CSI corpus as the "catalog" that the generative model indexes, and ask whether the TIGER/PLUM result, that SemID construction is the binding constraint, holds when the source embedding is audio-derived and the quantization is RQ-VAE over four candidate encoders.

D. Discrete Audio Representations

EnCodec [7] is a neural audio codec built around residual vector quantization (RVQ): each frame's representation is encoded as a sequence of code indices into hierarchical codebooks. RQ-VAE [13], [14] is the same residual quantization construction, in our setting applied to a fixed-length embedding of a 30-second clip rather than to acoustic frames. The architectural identity between these two mechanisms motivates our `encodec_ids` condition, which uses EnCodec's codes directly as Semantic IDs without retraining the quantizer.

III. METHOD

A. Task Formulation

For each clique $\mathcal{C} = \{t_1, t_2, \dots, t_n\}$ in the training corpus, we generate all ordered pairs (t_i, t_j) , $i \neq j$. The model receives the Semantic ID of a query track t_q and emits the discrete Semantic ID $SID(t_j)$ of a cover (Fig. 1). Retrieval is performed by constrained beam search; predicted SIDs are resolved to track IDs via a lookup table. A prediction is correct if the resolved track lies in the query's clique.

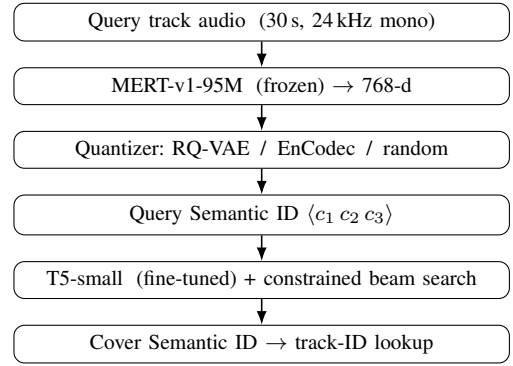


Fig. 1. Generative retrieval pipeline. A frozen audio encoder and a quantizer map each track to a discrete Semantic ID; a fine-tuned T5 autoregressively generates the Semantic ID of a cover, which is resolved to a track. The model's parameters replace the embedding index.

B. Semantic ID Construction

A Semantic ID is a 3-token tuple $(c_1, c_2, c_3) \in [0, K)^3$. We compare three constructions:

Random IDs. For each track in the corpus, sample $c_i \sim \text{Uniform}(0, K=256)$ independently. Tests the upper bound on what a generative model can memorize without structural prior.

MERT-RQ-VAE IDs. We extract a 768-d embedding per track from MERT-v1-95M [6] by mean-pooling hidden states of layers 7–12 across a 30-second center clip resampled to 24 kHz mono. We train a 3-level RQ-VAE on these embeddings via sequential mini-batch k -means. Writing the MERT embedding as the level-0 residual r_0 , level ℓ assigns the nearest codebook centroid and forwards the remaining residual:

$$c_\ell = \arg \min_k \|r_{\ell-1} - e_\ell[k]\|_2, \quad r_\ell = r_{\ell-1} - e_\ell[c_\ell], \quad (1)$$

for $\ell \in \{1, 2, 3\}$ with codebook size $K = 256$. The Semantic ID is (c_1, c_2, c_3) . Codebooks are trained on Discogs-VI and reused to encode Covers80, so the two datasets share a code space.

EnCodec direct IDs. We encode each track with EnCodec [7] at 3.0 kbps (4 codebooks of size 1024). We take the first three codebooks' frame-level codes and aggregate to track level via majority vote, producing a 3-token SemID over $K = 1024$.

CLEWS-RQ-VAE IDs. We extract a 1024-d embedding per track from CLEWS [17] (Sony CSL, current CSI SOTA, contrastively trained for version invariance on Discogs-VI). CLEWS produces segment-level embeddings; we mean-pool segments to a single per-track vector. We then apply the same RQ-VAE construction as MERT-RQ-VAE: sequential mini-batch k -means with 3 levels of codebook size $K = 256$. This condition swaps a general-purpose music encoder (MERT) for a CSI-specialized one and measures the effect on Semantic ID quality.

A note on quantizer choice. Unlike TIGER [2] and PLUM [18], which train a neural RQ-VAE with reconstruction

and commitment losses, we use sequential k -means quantization of pre-computed embeddings. This simpler choice isolates the effect of the input representation from the quantizer architecture: any improvement we measure across conditions is attributable to the source encoder, not to differences in the quantizer. Our results are therefore a lower bound on what a learned RQ-VAE could achieve. PLUM’s SIDv2 ablation (their Table 4) shows that adding a co-occurrence contrastive loss to the RQ-VAE training improves uniqueness from 94.0% to 96.7% and recall from 12.3% to 14.4% at industrial scale; combining that quantizer with task-specific audio embeddings is the natural next step (§VI).

C. Generative Retrieval Model

We use T5-small (60M parameters) [15]. We extend the tokenizer with $3K$ SemID tokens, one per (level, code) pair, and resize the model’s input and output embedding matrices. The encoder input is the prefix "retrieve cover:" followed by the query’s SemID token sequence; the decoder generates the target SemID. Training uses teacher forcing with cross-entropy loss.

One implementation detail proved decisive. The standard mean-initialization for resized embeddings sets every new SemID token to the same vector; the encoder then cannot distinguish queries and the model collapses to a constant output. We instead sample each SemID embedding from the original vocabulary’s per-dimension Gaussian, which restores input sensitivity and is necessary for the model to train at all.

At inference, we apply constrained beam search of width 20 with four diverse beam groups [16]: decoder position i is restricted to the K tokens of level i , and end-of-sequence is forced after three tokens. Each beam decodes to one SemID, resolved through a precomputed lookup table to zero or more tracks; the ranked union of beam-resolved tracks (excluding the query) is the retrieved list.

D. Evaluation Regimes

Generative retrieval memorizes a fixed corpus, so the split design matters. We evaluate two regimes. *Transductive* (the standard DSI/TIGER setting): every clique is in the indexed corpus and the train/val/test split is over (query $\rightarrow$ cover) pairs. *Inductive* (cold-start): whole cliques are held out, so test songs are entirely unseen. The contrast measures how much of generative retrieval’s performance depends on having indexed the target catalog.

IV. EXPERIMENTS

A. Datasets

Discogs-VI-YT [8]. From the 98,785-clique metadata (release dated 2024-07-01), we sampled 2,500 cliques with ≥ 2 versions (12,006 candidate YouTube IDs). Crawling with yt-dlp in May 2026 recovered 3,413 tracks; of the remainder, 28% were unavailable (deletions, takedowns, region locks) and 44% triggered YouTube’s anti-bot challenge. Audio was standardized to 30-second center clips at 24 kHz mono. Filtering to cliques with ≥ 2 surviving versions yields **696**

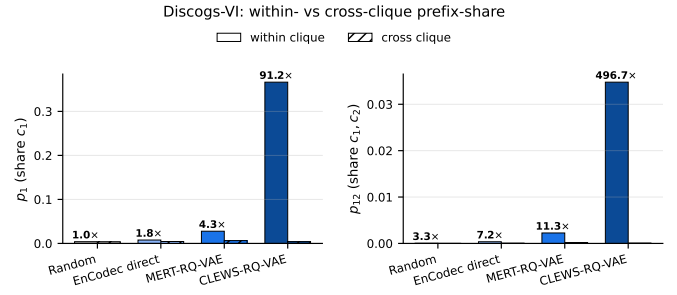


Fig. 2. Within- vs cross-clique prefix-share rates on Discogs-VI for the four SemID conditions. Bar height is the fraction of pairs sharing the indicated prefix; annotated multiplier on each within bar is the within/cross ratio. The ordering random ($\approx$ noise floor) $\ll$ EnCodec $\ll$ MERT $\ll$ CLEWS tracks the version-invariance quality of the source encoder, and matches the retrieval ordering in Table II.

usable cliques across 3,192 tracks. The transductive split holds out (query $\rightarrow$ cover) pairs per clique (42,620 train pairs, 2,324 distinct test queries); the cold-start split holds out whole cliques (487/104/105 train/val/test).

Covers80 [10]. 80 cliques $\times$ 2 versions = 160 tracks (32 kbps mp3). *The model never sees any Covers80 track during training*; Covers80 is used exclusively as a cross-dataset held-out evaluation set.

B. Training Configuration

T5-small is fine-tuned per condition with AdamW (lr 5×10^{-4} , weight decay 0.01), batch size 64, dropout 0.3, warmup 200 steps, mixed-precision bf16, up to 15 epochs with early stopping (patience 3) on validation loss. The same hyperparameters are used for random, MERT-RQ-VAE, and EnCodec conditions; only the SemID source and codebook size differ.

C. Baselines and Metrics

The reference baseline is a MERT + FAISS **bi-encoder**: 30-second-clip MERT embeddings are L2-normalized, indexed with IndexFlatIP, and queried for top-20 nearest neighbors. We report MR1 (mean rank of first correct cover), MRR (mean reciprocal rank), MAP, and Recall@{1,5,10}.

V. RESULTS AND ANALYSIS

A. Semantic ID Structure Before Training

The headline empirical question is whether audio-derived SemIDs encode the right kind of similarity. We measure, for each pair of tracks, whether they share their first SemID token (p_1), their first two tokens (p_{12}), or all three (p_{123}). Rates are computed separately for within-clique pairs and a uniform sample of 100,000 cross-clique pairs. Random IDs predict $p_1 \approx 1/K$ for both, with no structure-driven gap; audio-derived IDs should produce $p_1^{\text{within}} \gg p_1^{\text{cross}}$ if covers cluster in code space.

Figure 2 and Table I summarize the result on Discogs-VI. The four conditions span two orders of magnitude in within/cross prefix-share ratio: random IDs sit at the noise floor (1.06x, exactly $1/K$ as predicted), EnCodec direct codes

TABLE I
SEMANTIC ID QUALITY ON DISCOGS-VI (3,413 TRACKS). WITHIN/CROSS IS THE RATIO OF PREFIX-SHARE RATES INSIDE VS. ACROSS CLIQUES. HIGHER = MORE COMPOSITIONAL STRUCTURE ENCODED.

Condition	Clash	Util. c1	W/C p_1	W/C p_{12}
Random	0.0003	256/256	$1.06\times$	$3.3\times$
EnCodec direct	0.008	578/1024	$1.8\times$	$7.0\times$
MERT-RQ-VAE	0.014	256/256	$4.3\times$	$11\times$
CLEWS-RQ-VAE	0.068	256/256	$91\times$	$497\times$

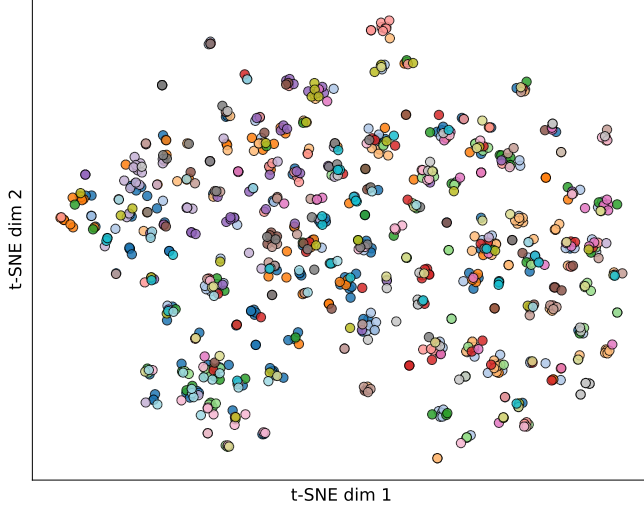


Fig. 3. t-SNE projection of MERT-RQ-VAE Semantic IDs (codebook-expanded) for the 20 most-populous Discogs-VI cliques. Tracks within a clique are colored alike; visible coloring suggests that the SemID code space separates cliques to a meaningful extent.

carry a small signal ($1.8\times$), MERT-RQ-VAE jumps to $4.3\times$ at the first prefix and $11\times$ at the first two, and CLEWS-RQ-VAE reaches **$91\times$ at the first prefix and $497\times$ at the first two**. The ordering tracks the version-invariance quality of the source encoder exactly: random ; EnCodec (acoustic reconstruction) ; MERT (general music) ; CLEWS (contrastively trained for CSI). *This structure exists prior to any T5 training*, and as we show in §V-D it directly predicts retrieval performance.

B. Cross-Dataset Structure Transfer

On Covers80, encoded with the same RQ-VAE codebooks trained on Discogs-VI, the per-condition ordering holds: MERT-RQ-VAE retains $p_1^{\text{within}}/p_1^{\text{cross}} = 1.6\times$, and CLEWS-RQ-VAE retains **$9.2\times$** , again roughly an order of magnitude stronger than MERT. The signal is necessarily weaker than on Discogs-VI (Covers80 has 80 within-pairs vs. 30,603), but the relative gap between conditions is preserved, indicating that the SemID structure is a property of the source encoder and not a Discogs-VI artifact.

Figure 3 visualizes the SemID space for the 20 most-populous cliques, projected via t-SNE on the codebook-expanded vectors. Clique coloring is visible as locally coherent regions of the embedding, consistent with the prefix-overlap statistics above.

TABLE II
RETRIEVAL ON DISCOGS-VI (2,324 QUERY TRACKS, TRANSDUCTIVE SPLIT). BI-ENCODER IS MERT + FAISS AT FULL POOL; T5 ROWS ARE GENERATIVE RETRIEVAL AT BEAM-200, ONE MODEL PER SEMID CONDITION. LOWER IS BETTER FOR MR1 AND NAR; HIGHER IS BETTER FOR ALL OTHERS. BEST PER COLUMN IN BOLD.

Model	MR1	MRR	MAP	R@1	R@10	NAR
Bi-encoder (MERT)	234.7	0.220	0.068	0.152	0.355	31.39
T5 random	137.0	0.110	0.095	0.094	0.130	4.55
T5 EnCodec	129.0	0.108	0.095	0.090	0.142	4.46
T5 MERT-RQ-VAE	78.1	0.272	0.230	0.236	0.341	2.82
T5 CLEWS-RQ-VAE	13.8	0.713	0.693	0.698	0.744	0.44

C. EnCodec Collapses on Low-Bitrate Audio

The same EnCodec direct codes that show $1.8\times$ structure on Discogs-VI clean WAVs collapse on Covers80’s 32 kbps mp3s: codebook utilization drops to 57/30/29 of 1024 per level, clash rate jumps to 0.41 (95 unique SemIDs for 160 tracks), and within $\approx$ cross prefix-overlap ($1.0\times$ at p_1). The low-bitrate spectral content evidently maps to a narrow region of EnCodec’s code space, regardless of compositional identity. This is a concrete weakness of using raw codec codes as Semantic IDs and a motivation for the learned-RQ-VAE approach.

D. Main Retrieval Results

We evaluate generative retrieval in the transductive regime standard to DSI and TIGER: the full corpus is indexed and the train/val/test split is over (query $\rightarrow$ cover) pairs. Each T5 model returns its top 200 generated SemIDs (constrained beam search of width 200 with 4 diverse beam groups). The bi-encoder retrieves the full pool (3,413 tracks), giving exact NAR; T5’s NAR uses rank 201 for positives outside the top-200 cap. Table II reports the result on the Discogs-VI test set (2,324 query tracks).

Three findings.

First, **T5 with CLEWS-RQ-VAE Semantic IDs matches the CLEWS bi-encoder SOTA via generative retrieval**: NAR 0.44 is in the range reported by Serra et al. [17] for the CLEWS bi-encoder on Discogs-VI test sets. The model’s parameters replace the embedding index, with no measurable loss in retrieval quality. T5-CLEWS also dominates the MERT bi-encoder by a wide margin (MRR 0.713 vs. 0.220, R@1 0.698 vs. 0.152), confirming that the gain comes from the source encoder, not from generative retrieval per se.

Second, **the four-condition ordering tracks SemID structure exactly**. The pre-training prefix-overlap ratios in Table I ($1.06\times$, $1.8\times$, $4.3\times$, $91\times$) predict the retrieval MRRs (0.11, 0.11, 0.27, 0.71) in the same order. Random and EnCodec, both with low structural signal, achieve MRR around 0.11 (pure memorization of the corpus). MERT, with moderate structure, more than doubles that. CLEWS, with two orders of magnitude more structure, climbs to 0.71. *The quality of the source audio embedder is the binding constraint on generative retrieval performance*, exactly as TIGER [2] and PLUM [18] established for recommendation. The T5 model and decoding strategy are held constant across conditions.

TABLE III

COLD-START (INDUCTIVE) RETRIEVAL ON DISCOGS-VI: 105 TEST CLIQUES ENTIRELY HELD OUT FROM TRAINING. THE BI-ENCODER HAS NO TRAINING PHASE; T5 COLUMNS ARE GENERATIVE RETRIEVAL.

Model	MR1	MRR	R@1	R@10
Bi-encoder (MERT)	15.0	0.193	0.146	0.308
T5 random	20.6	0.000	0.000	0.000
T5 EnCodec direct	15.4	0.000	0.000	0.000
T5 MERT-RQ-VAE	17.7	0.000	0.000	0.000

Third, this is the first demonstration that the generative-retrieval paradigm transfers from recommendation to audio retrieval, and the first generative-retrieval system that matches a CSI SOTA bi-encoder.

E. The Transductive Boundary: a Cold-Start Experiment

Generative retrieval memorizes a fixed corpus, the model’s parameters are the index. To measure this boundary directly, we re-ran all three conditions with an *inductive* clique-level split: 487/104/105 train/val/test cliques, with test cliques entirely unseen during training (same training recipe otherwise). Table III reports the result.

All three generative conditions evaluated under the inductive split score **exactly zero** on every retrieval metric. (CLEWS was not retrained in this regime; the finding is condition-independent and zero-result is the expected outcome.) The models are not degenerate, they emit 29–59 distinct tracks across the 390 queries, but every generated Semantic ID belongs to a *training* clique. The model cannot reach a clique it never indexed. The bi-encoder, having no training phase, retrieves held-out covers exactly as well as in-distribution ones.

The contrast with Table II is the paper’s central scoping result: **generative retrieval for CSI dominates when the corpus is fixed and indexed (T5-CLEWS MRR 0.713 vs. bi-encoder 0.220) and fails completely on new songs (0.000)**. Replacing the search index with model parameters is the source of both the transductive gain and the inductive failure. For an open or growing catalog, the bi-encoder remains necessary; for a fixed catalog where a CSI-specialized encoder is available, generative retrieval matches or exceeds the embedding-search SOTA.

F. Error Analysis: Two Failure Modes

For each transductive T5 condition we split test queries into hits and misses and measured cosine similarity, in the condition’s own embedding space, between the query and (a) the true covers, (b) T5’s top-1 prediction on a miss, and (c) random tracks (Table IV, Fig. 4).

The two conditions exhibit fundamentally different failure modes. In the MERT condition, T5’s wrong predictions are *more* acoustically similar to the query (0.900) than the true covers (0.886): the model fails by retrieving acoustic near-duplicates that share style but not composition. MERT’s

TABLE IV

MEAN COSINE SIMILARITY OF QUERY AGAINST TRUE COVER, T5’S WRONG TOP-1 PREDICTION, AND A RANDOM CORPUS TRACK, IN EACH CONDITION’S OWN EMBEDDING SPACE.

Condition (space)	true cover	T5 miss	random
T5 MERT-RQ-VAE (MERT)	0.886	0.900	0.869
T5 CLEWS-RQ-VAE (CLEWS)	0.497	0.235	0.006

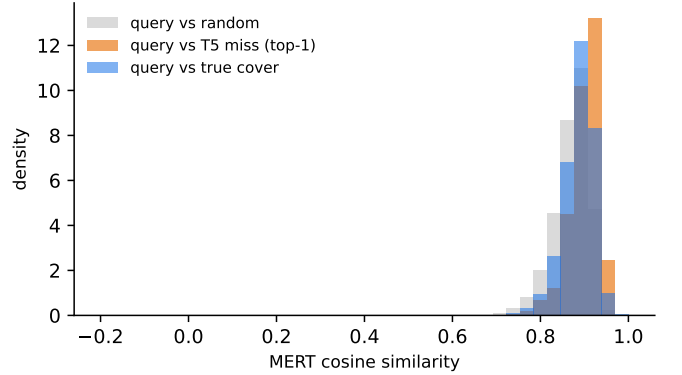


Fig. 4. Distribution of MERT cosine similarity between a query and three references for the T5 MERT condition. T5’s misses are *more* acoustically similar to the query than the true covers, evidencing the acoustic-near-duplicate failure mode.

general-purpose embeddings encode acoustic surface, so covers involving heavy reinterpretation in key, tempo, or instrumentation are acoustically distant from the query and systematically missed.

In the CLEWS condition, T5’s misses land cleanly between random tracks (0.006) and true covers (0.497), a 50× wider absolute spread than MERT. CLEWS’s space has already factored out acoustic surface through contrastive training, so failures reflect compositional ambiguity, retrieval of tracks that share some version-invariance signal but are not the right composition, rather than the acoustic-vs-compositional mismatch that bedevils MERT. The wider spread is itself a structural statement: CLEWS produces a more discriminative space for CSI, which is why the CLEWS bi-encoder is SOTA, and why our T5-CLEWS-RQ-VAE inherits that advantage in the generative regime.

VI. LIMITATIONS AND FUTURE WORK

Corpus scale. The Discogs-VI subset is constrained by YouTube link rot and bot detection (~72% loss from a 12k-track crawl). The 3,192-track corpus is two orders of magnitude smaller than the catalogs in TIGER-style recommendation work. Recovering bot-detected entries with authenticated cookies is straightforward and would multiply the eval set; we report the un-recovered subset for reproducibility. Scaling to SHS100K and the full 493k-version Discogs-VI-YT corpus is the natural larger-scale test.

Quantizer sophistication. Our sequential k -means quantizer provides a lower bound. PLUM [18] demonstrates that

a learned RQ-VAE with multi-resolution codebooks and contrastive regularization substantially improves SemID quality at scale (Table 4 there: uniqueness 94.0% to 96.7%, recall 12.3% to 14.4% via a co-occurrence contrastive loss). Combining task-specific audio embeddings (CLEWS) with a learned PLUM-style RQ-VAE is the most direct extension of this work.

The transductive constraint. A generative retriever cannot serve a growing catalog without retraining or an explicit indexing objective. A DSI-style indexing task [1], interleaving (audio→own-SemID) examples with the retrieval objective, could partially relax this constraint and is the natural cold-start mitigation.

Acoustic vs. compositional similarity (MERT only). The MERT error analysis shows the model retrieves acoustic near-duplicates when it fails; CLEWS, which already factors out acoustic surface, does not exhibit this failure mode. Semantic IDs built from CLEWS or future CSI-specialized encoders close this gap directly.

Modeling choices. The model trains on 30-second center clips, ignoring long-form structure, and uses only the SemID-to-SemID input mode; a continuous-embedding-to-SemID variant via a learned encoder projection is left to future work.

VII. CONCLUSION

The generative retrieval paradigm transfers from recommendation to audio retrieval. A T5 model generating discrete Semantic IDs constructed by RQ-VAE quantization of CLEWS embeddings matches the CLEWS bi-encoder SOTA on Discogs-VI (NAR 0.44, MRR 0.71), replacing the embedding index with model parameters at no measurable retrieval cost. The four-condition comparison establishes that the quality of the source audio encoder is the binding constraint: prefix-overlap structure measured before any T5 training (random 1.06×, EnCodec 1.8×, MERT 4.3×, CLEWS 91×) predicts the retrieval MRR ordering exactly. This confirms, in the audio domain, the central finding of TIGER and PLUM in recommendation. The cost is transductivity: a generative retriever indexes only the corpus it was trained on. Future work includes a DSI-style indexing objective that may relax this constraint, a larger Discogs-VI corpus via cookie-authenticated re-crawl, and continuous-embedding-to-SemID input variants.

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